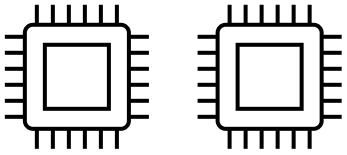
NOVA SCHOOL OF SCIENCE & TECHNOLOGY

Circuits and Systems for Radio-Frequency

Communication Link Simulation in GNU-Radio
Testing of a Low-IF receiver (SDR)



Contact:
João P. Oliveira
jpao@fct.unl.pt

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Scientific area: Electronics
Department of Electrical and Computer Engineering
Nova School of Science & Technology – NOVA University of Lisbon

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1- Introduction

This lab intends to introduce some basic concepts associated with radio communication link, radio transmitters, and receivers. Learning Objectives:

- Understand basic AM communication link concepts.
- Practice GNU Radio system-level simulation.
- Use SPICE-based modeling of AM systems.
- Design and simulate an RF band-pass filter
- Perform practical experiments with a VNA and RFFE.
- Compare theoretical, ideal-simulation, realistic-component simulation, and measured results.

The lab consists of three parts:

1. Gnuradio-companion AM system simulation,
2. Spice-based AM electrical model simulation,
3. Design, simulation and experimental characterization of an FM-band RF filter, followed by experiments with a VNA and a Radio Frequency Front-End (RFFE).

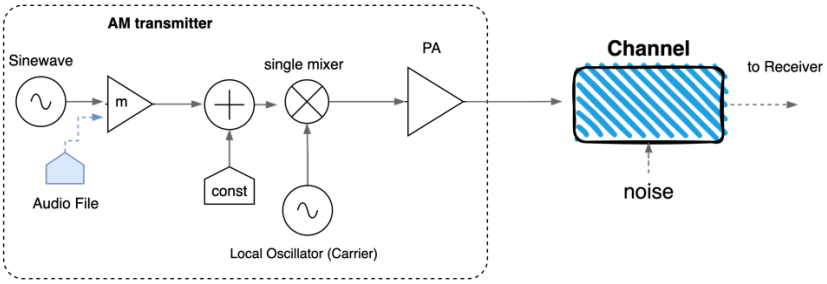
The work is intended to be done by groups of three students. A total of three weeks of lab classes are allocated to the lab assignment. A final report must be delivered by the end of the last week.

The subjects and tools addressed in this lab are described in:

- Chapter 2 to 4 from Prof. B. Razavi book, "RF Microelectronics, 2nd Edition"
- Gnuradio companion https://wiki.gnuradio.org/index.php/Main_Page
- LTspice, https://www.analog.com/en/design-center/design-tools-and-calculators/ltspice-simulator.html
- QUCS-S: https://ra3xdh.github.io/
- QUCSstudio : https://qucsstudio.de/

2- AM communication link simulation in GNU-RADIO

Figure 1 shows the simplified block diagram of the system to be simulated on gnuradio-companion. It consists of an AM transmitter, a channel block model, and an AM receiver.



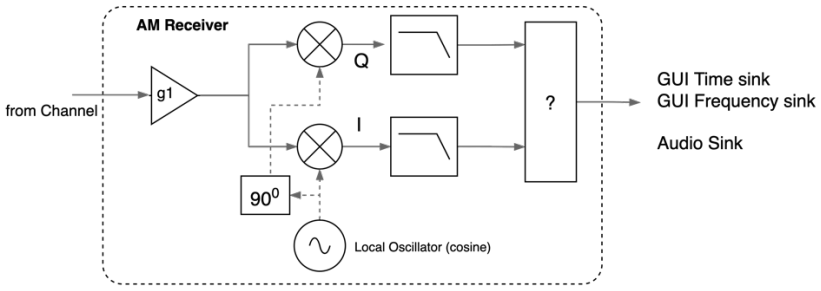


Figure 1 - Simplified block diagram of the AM system to be simulated on Gnuradio companion.

Draw the complete block diagram that models the AM communication system. You can start with the simplified diagram of Figure 1. However, additional blocks can/must be added to improve the model. For example, a distortion block can be added to simulate the effects that originates the 1dB compression, and other non-linearities. It is mandatory to use a quadrature mixer (IQ) on the receiver side. On the transmitter, you can choose any option.

Consider the following suggestions and remarks:

- Carrier frequency between 50kHz and 100kHz. For the sampling, a value of 768kHz may be used, but another one can be selected depending on your analysis;
- Build channel model block. Add a QT GUI range to change the noise level;
- Use multipliers and adders to model the mixing operation;
- Each time a float to complex, or vice-versa, conversion use Float-to-Complex and Complex-to-Float blocks;
- Include QT GUI Time Sink blocks to visualize signals in the time domain;
- Include QT GUI Frequency sink blocks to visualize signals in the frequency domain;
- As input signal, use a Signal Source with cosine waveform (begin with 1kHz of frequency);
- After the complete model is successfully tested with this ideal signal, use a 48ksps wave audio file as a source, employing a Wav File Source block. To listen the audio, use an Audio Sink block with sample rate of 48ksps (depending on the sound card of your computer);
- When running gnuradio-companion in simulation mode only (i.e., no hardware involved), use the Throttle block to limit the CPU usage;

Deliverables:

- Complete block diagram of the system in Gnu-Radio.
- Screenshots/plots of time and frequency signals.
- Discussion on SNR and effect of channel noise.
- Optional: BER analysis when using digital test signals.

3- AM communication link simulation in spice

Develop a suitable SPICE-based electrical model for the complete AM communication link, including transmitter, channel and receiver. The circuit structure and the implementation of the required functional blocks should be selected and justified by the group. In the receiver, it is not mandatory to use I/Q mixing. Use circuit-level models for the amplification, mixing and filtering functions. At this stage, ideal or simplified component models may be used. After setting up the system/circuit schematics, obtain the simulation results (in time and frequency) that demonstrate the transformation undergone by the signal along the channel path (transmitter, channel, and receiver).

Consider the following suggestions and remarks:

- Carrier frequency between 50kHz and 100kHz;
- Use “ideal” switches to model the mixing operation; The LO is a square wave with an amplitude of 3 V (for example);
- Observe the signal in time and frequency (through FFT) at several points in the signal path;
- As input signal, use a Signal Source with cosine waveform (begin with 1kHz);
- LTspice, Qucs-s, QucsStudio, or other spice based simulators can be used.

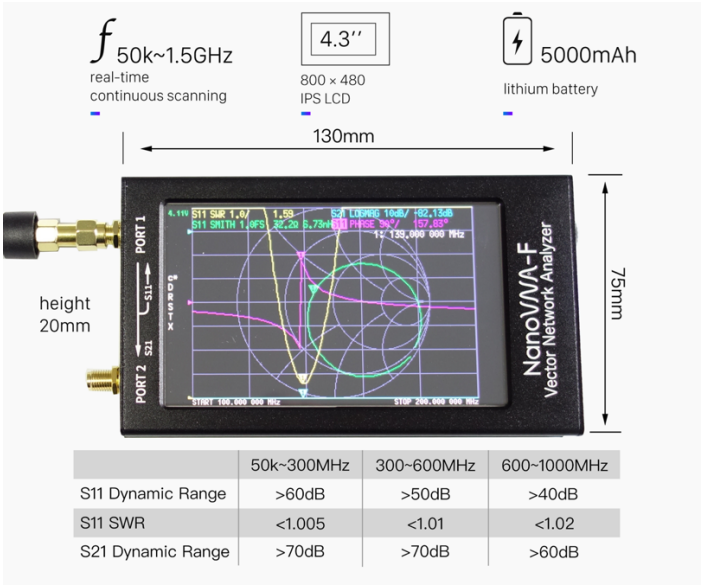
Deliverables:

- System schematic in SPICE.
- Time and frequency domain plots at key nodes.
- Comparison with GNU Radio results.

4- Lab Experiments with an RF Filter, VNA and RFFE

4.1 - 2nd-order FM-band band-pass filter

This part of the lab addresses the design, simulation and eventual experimental characterization of a simple passive 2nd-order RF band-pass filter for the FM broadcast band. The filter topology is fixed and consists of two LC resonators: a series LC resonator in the signal path followed by a parallel LC resonator connected to ground. Each group will be assigned a centre frequency in the FM band (nominally around 94 MHz; different centre frequencies may be assigned to different groups). The filter is designed for 50 Ω source and load impedances and will later be inserted in the receiver front-end described in Section 4.2. The nanoVNA shown in Figure 2 can be used to measure the filter S-parameters.



	50k~300MHz	300~600MHz	600~1000MHz
S11 Dynamic Range	>60dB	>50dB	>40dB
S11 SWR	<1.005	<1.01	<1.02
S21 Dynamic Range	>70dB	>70dB	>60dB

Figure 2 - Low-Cost Vector Network Analyzer.

Use the specified 2nd-order two-resonator topology and determine suitable component values for the centre frequency assigned to your group. The objective is a relatively broad RF preselector rather than a channel-select filter. A bandwidth of approximately 15–25 MHz is a reasonable design target. Begin with an ideal-component design and then evaluate the impact of real component behaviour at RF. The implementation should use SMD components, preferably 0805 (0603 may be used when necessary), selected only from the set of values supplied in the laboratory. Capacitors should preferably use a C0G/NP0 dielectric, and RF inductors from the selected component family should be used whenever possible.

a) For the centre frequency assigned to your group, estimate initial LC values for the two resonators. Use these values as a starting point for the complete 50 Ω terminated network. The topology and order are fixed; the design task is to select the component values and obtain an adequate passband around the assigned frequency.
b) Simulate the complete filter initially using ideal components. Determine the frequency response and relevant parameters, including centre frequency, -3 dB bandwidth, insertion loss and input matching (S11). Optimize the design using only values from the component set available in the laboratory; arbitrary component values should not be used in the final design.
c) Replace the ideal inductors and capacitors by realistic models of the selected commercial components, using manufacturer models when available or a simple equivalent model including the most relevant losses and parasitics. No detailed component characterization is required. Verify briefly that the selected inductors and capacitors are suitable for operation around 100 MHz (in particular, adequate Q and self-resonant frequency) and compare the resulting filter response with the ideal-component simulation in terms of centre frequency, bandwidth, insertion loss and matching.
d) If the laboratory PCB is available, assemble and solder the filter using the selected SMD components.
e) Calibrate the nanoVNA and measure the filter S-parameters, at least S11 and S21, over a suitable frequency range covering the FM band. Extract the measured centre frequency, -3 dB

bandwidth and insertion loss. Compare the measured response with both simulations and discuss the main differences.

4.2 - Experiments with a RFFE

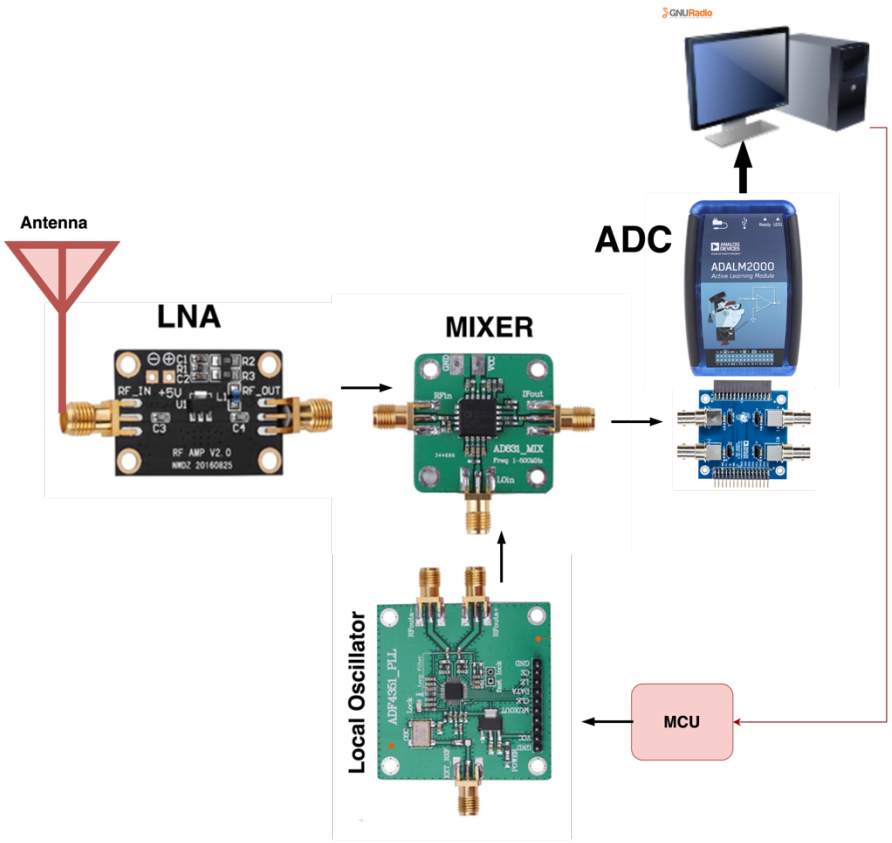


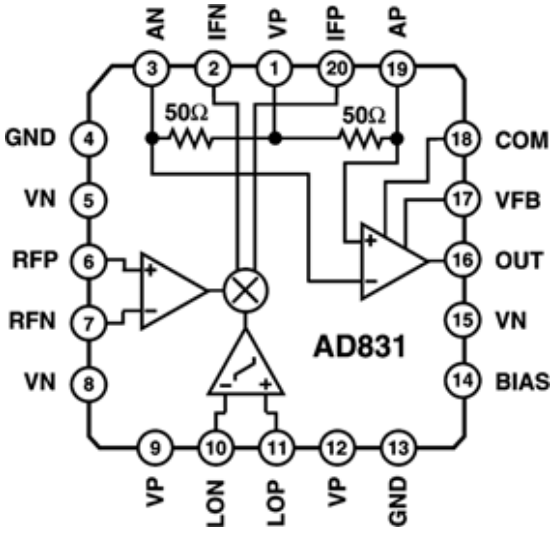
Figure 3 - Circuit blocks for the Radio-Frequency Front-End.

The objective of this experiment is to build and test a low-IF receiver able to receive FM radio (87.5 to 108 MHz). The key components of the front-end system are the FM-band band-pass filter developed in Section 4.1, a Low Noise Amplifier (LNA), a Mixer (AD831), and a Phase-Locked Loop (PLL) synthesizer (ADF4351). The system downconverts the selected RF signal to low-IF, which is further processed in GNU Radio after being digitized by the ADC of the ADALM2000 platform.

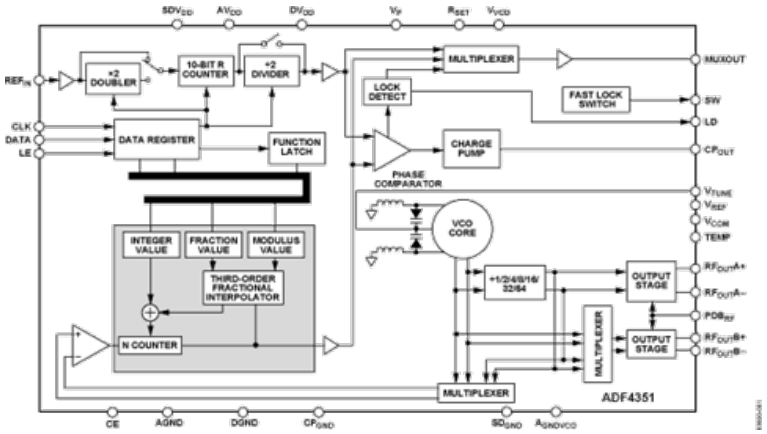
The RF front-end system consists of the following blocks:

1. FM-band band-pass filter: Provides RF preselection around the centre frequency assigned to the group. It is inserted between the antenna and the LNA. Its insertion loss and measured response should be considered when analyzing the complete receiver.
2. Low Noise Amplifier (LNA): Amplifies the incoming RF signal with minimal added noise, improving the system's overall sensitivity.
3. Mixer (AD831): The mixer combines the amplified RF signal with the LO signal generated by the PLL. The resulting output is a downconverted signal at a low IF.

4. Phase-Locked Loop (PLL) Synthesizer (ADF4351): The PLL generates a stable and programmable LO signal to drive the mixer. In this case, the LO will be set to produce a low-IF signal from the mixer.
5. The ADC is included in the ADALM2000 platform.
6. Gnu-Radio



The Mixer board is based on the AD 831 IC from analog devices
https://www.analog.com/en/products/ad831.html



The Local Oscillator board is based on the PLL ADF 4351 from Analog Devices
https://www.analog.com/en/products/adf4351.html

Receiver Assembly

1. Filter and LNA Setup: Connect the antenna to the input of the FM-band filter developed in Section 4.1 and connect the filter output to the LNA input using SMA coaxial cables. The LNA supply voltage is 5 V.

2. Mixer Setup: Connect the LNA output to the RF input of the mixer. The LO input of the mixer will be connected to the output of the PLL synthesizer (ADF4351), which will provide a frequency close to the RF frequency to produce the desired low IF. The IF output from the mixer connects to the spectrum analyzer or the ADALM200 for measurement. The supply voltage is 9V.

3. PLL Configuration: Select an FM broadcast station within the passband of the implemented RF filter. Program the PLL so that the selected RF signal is converted to a low IF (around 1 MHz). The LO may be placed below or above the RF signal; clearly indicate the selected frequency plan. Connect the PLL output to the LO input of the mixer using an SMA coaxial cable. The supply voltage is 5 V.

4. Intermediate Frequency (IF) Output: The IF output from the mixer is connected to the spectrum analyzer or the ADALM2000. This is the first downconverted signal step of the RF front end. The second mixing stage is performed in the digital domain, using Gnu-radio.

5. Gnu-Radio script. Use the given processing flow in Gnu-Radio. The block diagram includes the ADALM2000 block, a mixing stage, filtering and a FM radio demodulator and finally an audio-sink.

Receiver Testing

Turn on the power and test the setup by tuning to the selected radio-station frequency. When possible, compare the receiver operation with and without the RF band-pass filter and relate the observed behaviour to the measured filter response.

Deliverables:

- 2nd-order band-pass filter design calculations and selected component values from the laboratory set.
- Key ideal and realistic-component filter simulation results.
- Measured filter S-parameters (at least S11 and S21) and a concise comparison with simulation.
- Experimental integration of the filter in the RFFE and observations on its effect on receiver operation.
- RFFE chain gain/NF estimation.
- Brief description of the of PLL setup and tuning and mixer operation
- Comments on the FM demodulated audio.

5- Final Report Guidelines

The final report should be concise and must not exceed 8 pages, including figures, tables and references. Only the most relevant schematics, plots and measurements should be included; repeated screenshots and extensive raw data should be avoided. Appendices can be included but are not required. A suggested organization is:

1. Introduction and objectives
2. GNU Radio setup, main simulation results and comments
3. SPICE communication model: main results and analysis
4. RF band-pass filter: design, ideal/realistic simulation, VNA measurement and comparison
5. RFFE experiment: setup, filter integration and main results
6. Conclusions

References : consult moodle.